Term 3 answer key

Formulas printed for students in every question. Answers with π are rounded to 1 decimal place, as each question says; probabilities are fractions in simplest form; every other answer is exact.

WEEK	SESSION	Q1 A Angles & shapes	Q2 B Measurement	Q3 C Number & percentages	Q4 D Algebra, ratio & rates	Q5 E Number plane, data & chance
1	1	115°	42 cm	−13	$m = 18$	7/10
	2	65°	78 m²	0.4	07:15	1/6
	3	38°	24 cm²	$17	21	(2, 5)
2	1	95°	64 cm³	80%	$20 and $70	4
	2	115°	70 m²	1240	$4x - 12$	23
	3	133°	50 m²	36 m	4 km/h	3/5
3	1	125°	37.7 m	32	Bag A ($3 per kg)	6
	2	51°	26 cm	0.35	$x = 5$	4/5
	3	152°	750 mL	$16	24	1/5
4	1	124°	36 cm²	90%	4 : 7	5, 7
	2	93°	70 cm²	$60	$2x + 14$	2
	3	125°	56 m²	6	45 minutes	7/8
5	1	110°	80 cm³	0.09	$x = 7$	19 cm
	2	70°	36 cm²	72%	60 and 12	14
	3	44°	62.8 cm	$36	22	9/25
6	1	117°	60 cm	55%	5 km/h	1/2
	2	112°	54 cm²	12	$7y + 21$	3
	3	137°	30 cm²	18	$24 per hour	5/6
7	1	36°	7000 m	4/5	4 : 3	2, 8
	2	67°	42 cm²	48	$p = 6$	9
	3	116°	61 m²	43 kg	35	20
8	1	48°	100 m³	−5	5:25 pm	9/10
	2	124°	78.5 cm	35%	$5x + 10$	1/3
	3	66°	36 cm	$75	24 and 32	4/7
9	1	48°	56 m²	12%	5 km/h	(4, 1)
	2	48°	400 m²	21 kg	$x = 8$	4
	3	55°	80 cm²	125	16	9
10	1	100°	40 cm²	0.82	$27 per hour	7/25
	2	70°	450 mm	75%	2 : 3	6
	3	65°	25.1 cm	$56	$6x + 10$	7/12

A MA4-ANG-C-01 complementary and supplementary angles, angles at a point, vertically opposite angles, angles on parallel lines; MA4-GEO-C-01 angle sum of a triangle and a quadrilateral, exterior angle of a triangle

B MA4-LEN-C-01 perimeter, circumference, length units; MA4-ARE-C-01 area of rectangles, squares, triangles, parallelograms and composite shapes, area units; MA4-VOL-C-01 volume of rectangular prisms, volume and capacity units

C MA4-INT-C-01 order of operations with integers; MA4-IND-C-01 powers, square and cube roots; MA4-FRC-C-01 fraction and percentage of a quantity, one quantity as a percentage of another, fractions, decimals and percentages, multiplying and dividing by 10, 100 and 1000*

D MA4-ALG-C-01 substitution, the distributive law; MA4-EQU-C-01 one- and two-step equations; MA4-RAT-C-01 simplifying ratios, sharing in a ratio, rates, best buys, speed; 24-hour time and time differences*

E MA4-LIN-C-01 tables of values, reading points on the number plane; MA4-DAT-C-01 dot plots; MA4-DAT-C-02 mean, median, mode and range; MA4-PRO-C-01 probability, complementary events, experimental probability*

Codes are from the NSW Mathematics K–10 Syllabus (2022). The NESA website could not be reached from the build machine, so **none of these codes was confirmed on the NESA site**; they match the codes used in the Year 9 and Year 10 keys.

* the topic is in this outcome as far as could be checked, but where it sits in the syllabus (or, for 24-hour time, whether it is Stage 3 revision with no Stage 4 code) could not be confirmed.